Mathematics III_2023_PYQ

Group-A (Very Short Answer Type Questions)

Answer any ten of the following: (1 × 10 = 10)

1.

a) What is the area of the region bounded by x-axis, $y = e^x,\ x = 0\ , x = 1$?

b) What is the general form of Clairaut's equation?

c) If a graph has 5 vertices and 7 edges, then what is the size of its adjacency matrix?

d) On which region $\log(1 + x)$ can be expanded in an infinite series?

e) If for any $\vec{A}$, $\nabla \times \vec{A} = 0$, then $\vec{A}$ will be called as?

f) Find the value of $\int_{x=1}^{x=-1} \int_{y=2}^{y=-2} \int_{z=3}^{z=-3} x\, y^2 z^3 dz\ \ dy\ dx$

g) $\int_C y\, dx + x dy = p$, where C is given by $x = \cos\theta$, $y = \sin\theta$, $0 \le \theta \le \pi/2$. Find value of p.

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h) Find the value of $\frac{1}{D^2+4}(\sin 2x)$.

i) What is the eccentricity of the vertex of a graph having only one vertex?

j) What is the nature of the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \ldots$?

k) If $f(x, y) = |x| + |y|$, find the value of $f_x(0,0)$?

l) If C is the circle $x^2 + y^2 = 4$, find the value of $\int_C x^2\, dx$.

Group-B (Short Answer Type Question)

Answer any three of the following: (5 × 3 = 15)

2. Test the series $\sum_{n=1}^{\infty} (\frac{n}{(n+1)}) n^2$.

3. If $z = u^2 + v^3$, where $u = \sin xy$ and $v = y^2$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

4. Verify that $e^{(\tan^{-1} x)} = 1 + x + \frac{x^2}{2} - \frac{x^{-3}}{6} - \ldots$

5. Find $\frac{dy}{dx}$ of the function $(\sin y)^x - (\cos x)^y = 0$.

6. Find the general and singular solution of $y = 4xp - 16y^3p^2$, where $p = \frac{dy}{dx}$.

Group-C (Long Answer Type Question)

Answer any three of the following: (15 × 3 = 45)

7. (a) Test the convergence of the series whose nth term = $\left(n^{1/n} - 1\right)^n$.

   (b) Examine the convergence of the series $\frac{1}{a+b} + \frac{1}{a+2b} + \frac{1}{a+3b} + \cdots (a > 0, b > 0)$.

   (c) Assuming the validity of expansion, show that

$$\sin\ x = 1 - \frac{(x - \frac{\pi}{2})^2}{2!} + \frac{(x - \frac{\pi}{2})^4}{4!} - \cdots$$

8. (a) If $u = \log r$ and $r^2 = x^2 + y^2 + z^2$, prove that

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$$r^2 \frac{\partial^2 u}{\partial x^2} + r^2 \frac{\partial^2 u}{\partial y^2} + r^2 \frac{\partial^2 u}{\partial z^2} = 1$$

(b) Show that $f(x, y) = 3x^3 + 4x^2y - 3xy^2 - 4y$ is neither a maximum nor a minimum at (0, 0).

(c) Determine the constant m so that the vector $\vec{v} = (x + 3y)\hat{\imath} + (y - 2z)\hat{\jmath} + (x + mz)\hat{k}$ is solenoidal.

9. (a) If $u_n = \frac{3^n}{n+1}$, show that $[u_n]$is monotonic increasing and bounded above, find its limit.

   (b) Expand e^x in power series of $(x - 1)$.

   (c) Examine the convergence of the series $\sum u_n$, where $u_n = \frac{(n+1)(n+4)}{n(n+2)(n+5)}$.

10. (a) If $u(x, y) = f(x + 2yz, y^2 + 2zx)$, prove that

$$(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$$

(b) If $u = \tan -1\left(\frac{x5/2+y^{5/2}}{\sqrt{x}-\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.

(c) Show that the function $f(x, y) = 4x^2y - y^2 - 8x^4$ has a maximum value at (0, 0).

11. (a) The given function $f(x,y) = \frac{xy(x2-y2)}{x2+y2}, (x,y) \neq (0,0) = 0, (x,y) = (0,0)$. Find from definition $f_{xy}(0,0)$ and $f_{yx}(0,0)$.

(b) If $A = \pi h^2 \frac{\sin\alpha}{1-\sin\alpha}$, find dA where h and α are independent variables.

(c) If $f(x,y) = \frac{x+y}{1-xy}$ and $g(x,y) = \tan^{-1} x + \tan^{-1} y$, find the Jacobian $\frac{\partial(f,g)}{\partial(x,y)}$.

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